

MARKET REGIME DETECTION & SWITCHING RISK

ATISH MOTIRAM SAWANT

Roll No. 242123104



**DEPARTMENT OF MATHEMATICS
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI, INDIA**

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What Are Financial Markets?

Definition

Financial markets are platforms where financial assets such as stocks, bonds, and securities are bought and sold

Why Study Financial Markets?

- Financial markets drive the global economy
- Prices evolve continuously based on new information

Key Question

Are markets predictable or completely random?

- Markets show **uncertainty and randomness**
- BUT patterns exist in **returns, volatility, and risk**

What is Volatility?

Simple Idea:

Volatility measures how much prices move up and down

Two scenarios:

- **Low Volatility** → Smooth and stable market, small price changes
- **High Volatility** → Uncertain and risky market, large price swings

Key Insight

High volatility = higher risk

Do Markets Really Behave Randomly?

- Prices appear random in the short term
- But real data shows **clear patterns over time**
- **Key observation:** Markets do not behave the same all the time
- We see periods of:
 - Stable growth
 - High volatility
 - Sudden crashes and recoveries

So can we identify these different phases?

What Are Market Regimes?

Definition

Market regimes are distinct phases where market behavior remains similar for a period of time

Examples of regimes:

- **Bull Regime** → High returns, low volatility
- **Normal Regime** → Moderate returns, moderate volatility
- **Crisis Regime** → Negative returns, high volatility

Key Challenge: Regimes are **not directly visible** — we must detect them from data

Why Do Investors Care?

- Investors are not only concerned with returns
- The biggest concern is **losses during bad markets**

Drawdown

A drawdown is the fall in value from a peak to a lower point

Example:

- Investment grows from 10,000 $\rightarrow$ 12,000
- Falls to 9,000
- **Loss = 25% from peak**

Important Insight:

Different market regimes lead to very different levels of risk

How Do Regimes Affect Risk?

Key Idea: Risk is NOT constant over time

- **Bull Regime**
 - Steady growth
 - Small and short-lived losses
- **Crisis Regime**
 - Sharp price declines
 - Large and prolonged losses

Important Insight

Same strategy behaves very differently in different regimes

Goal

Can we detect regimes early and adapt?

How Can We Solve This? (Role of AI)

Problem:

- Market regimes are **hidden**
- Cannot be directly observed from prices

Solution: Use Artificial Intelligence

- Detect patterns from historical data
- Capture relationships between:
 - Returns
 - Volatility
 - Risk measures
- Adapt to changing market conditions

Key Idea

AI helps uncover hidden market structures

Methods Used in This Project

Approach: Unsupervised Learning

Why Unsupervised?

We do not have labels for market regimes

Two Key Methods:

- **K-Means Clustering**
 - Groups similar market conditions
 - Provides initial segmentation
- **Hidden Markov Model (HMM)**
 - Captures time evolution of markets
 - Models transitions between regimes

Key Insight

K-Means finds structure, HMM captures how regimes change over time

Dataset Used

Data Source:

- S&P 500 Index (Benchmark US market index)
- Data obtained using **yfinance library**

Time Period:

- January 2001 → January 2025
- Around **24 years of daily data**

Why this dataset?

- Covers multiple market conditions:
 - Dot-com crash (early 2000s)
 - Global financial crisis (2008)
 - COVID-19 crash (2020)

Feature Engineering

Goal: Convert raw prices into meaningful signals

Key Features Used:

- **Returns** - Measures price change over time
- **Volatility** - Measures how much prices fluctuate
- **Drawdown** - Measures loss from peak value
- **Rolling Statistics** - Captures short and long-term trends

Key Idea

Features capture both market movement and risk

Final Dataset for Modeling

After preprocessing, we obtain:

- Each row = **one trading day**
- Each column = **a feature**

Feature Normalization

All features are scaled to ensure fair comparison

Ready for Modeling

Data is now suitable for AI algorithms

Dataset Preview

	Price	return	volatility	running_max	drawdown	drawdown_duration	rolling_mean_20	rolling_vol_20	rolling_mean_100	rolling_vol_100
Date										
2001-06-22	-0.736406	-0.806692	-0.118009	-0.798208	0.076784	-0.633422	-1.259531	-0.118009	-1.390794	0.658217
2001-06-25	-0.741733	-0.478415	-0.151136	-0.798208	0.035556	-0.631101	-1.124101	-0.151136	-1.389688	0.658158
2001-06-26	-0.743186	-0.146691	-0.164417	-0.798208	0.024317	-0.628780	-0.989712	-0.164417	-1.458629	0.655567
2001-06-27	-0.747677	-0.409266	-0.247921	-0.798208	-0.010437	-0.626460	-0.753609	-0.247921	-1.330393	0.640167
2001-06-28	-0.735735	1.004580	-0.188239	-0.798208	0.081975	-0.624139	-0.620928	-0.188239	-1.242840	0.649668

K-Means Clustering

Goal:

Group similar market conditions together

How it works:

- Looks at data points (features)
- Groups them into **K clusters**
- Points in same cluster behave similarly

In our case:

- $K = 3$ clusters
- Each cluster = a market regime (Bull, Normal, Crisis)

Key Idea

Similar patterns → Same market regime

K-Means Results

We obtained 3 clusters:

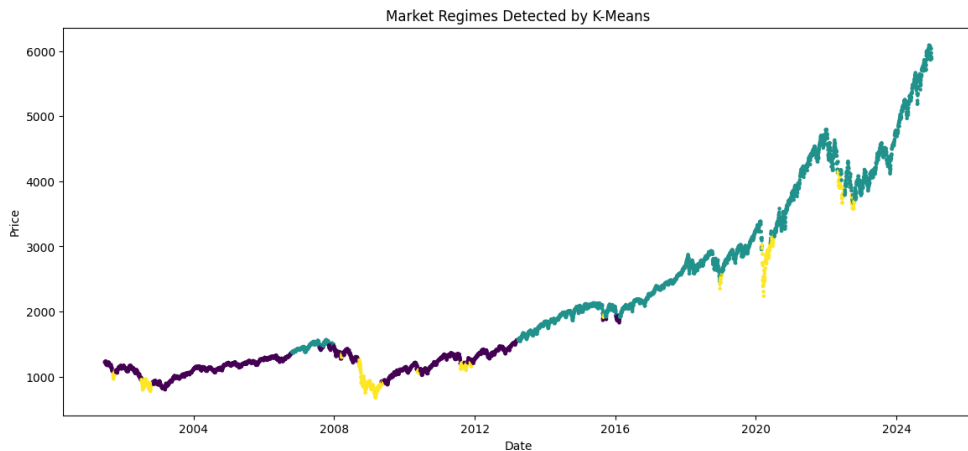
- **Regime 1: Bull Market**
 - Positive returns
 - Low volatility
 - Small drawdowns
- **Regime 0: Normal Market**
 - Moderate behavior
 - Balanced risk and return
- **Regime 2: Crisis Market**
 - Negative returns
 - High volatility
 - Large drawdowns

Key Insight

K-Means successfully separates market conditions

K-Means Regime Visualization

Market regimes identified using K-Means



Limitation of K-Means

K-Means gives useful grouping, but...

- It treats data as **independent points**
- It ignores the **time order** of markets

Problem

Markets evolve over time — regimes do not change randomly

Example:

- A crisis does not appear suddenly for one day
- It persists for a period of time

Solution

We need a model that captures **time dynamics**

Hidden Markov Model (HMM)

Main Idea: Markets move between hidden states over time

- Regimes are **not directly visible**
- We only observe returns and volatility
- HMM identifies these hidden states from data

Key Advantage

HMM captures **how markets evolve** and transition between regimes

Transition Between Regimes

- HMM calculates **transition probabilities** between regimes
- Our data shows: High probability of staying in same regime

Key Insight

Regimes are **persistent** — they last for extended periods

HMM Results: Market Regimes

HMM identifies 3 hidden regimes:

- **Regime 1: Bull Market**
 - Positive returns
 - Low volatility
 - Minimal drawdowns
- **Regime 0: Normal Market**
 - Moderate returns and risk
- **Regime 2: Crisis Market**
 - Negative returns
 - High volatility
 - Large drawdowns

Key Insight

HMM provides a more realistic view of market behavior over time

HMM Transition Matrix

Probability of moving between regimes

From ↓ To →	Normal (R0)	Bull (R1)	Crisis (R2)
Normal (R0)	0.98	0.01	0.00
Bull (R1)	0.02	0.98	0.00
Crisis (R2)	0.01	0.00	0.99

- Diagonal values are very high (0.98-0.99)
- Market tends to stay in the same regime
- Very low probability of switching

Key Insight

Market regimes are highly persistent

Regime Duration Analysis

How long does each regime last?

Regime	Mean Days	Median	Max Days
Normal	58	20	714
Bull	51	36	310
Crisis	117	47	607

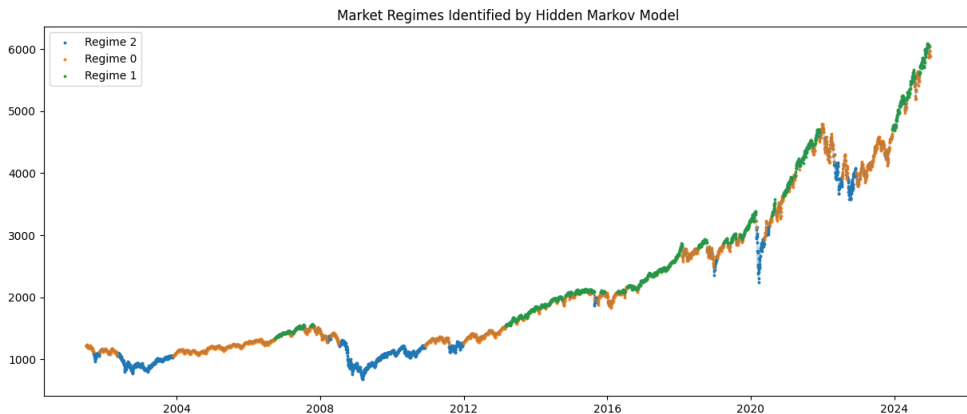
- Crisis regime lasts the longest on average
- Recovery periods are slow

Key Insight

Once a crisis starts, it tends to persist

HMM Regime Visualization

Market regimes identified using HMM



Using Regimes for Investment

Key Idea: Different market regimes require different strategies

Two approaches:

- **Buy-and-Hold Strategy**

- Always invested in the market

- **Regime-Based Strategy**

- Invest only in favorable regimes (Bull market)
- Stay in cash during risky periods (Normal & Crisis)

Main Question

Can regime-based investing improve performance?

Performance Evaluation Metrics

How do we compare strategies?

- **Total Return** - Overall growth of investment
- **Volatility** - Measures risk (price fluctuations)
- **Sharpe Ratio** - Return earned per unit of risk
- **Max Drawdown** - Worst loss from peak to trough

Sharpe Ratio Formula

$$\text{Sharpe Ratio} = \frac{\text{Mean Return}}{\text{Volatility}} \times \sqrt{252}$$

Higher Sharpe → better risk-adjusted performance

Performance Comparison

Buy & Hold vs Regime-Based Strategy

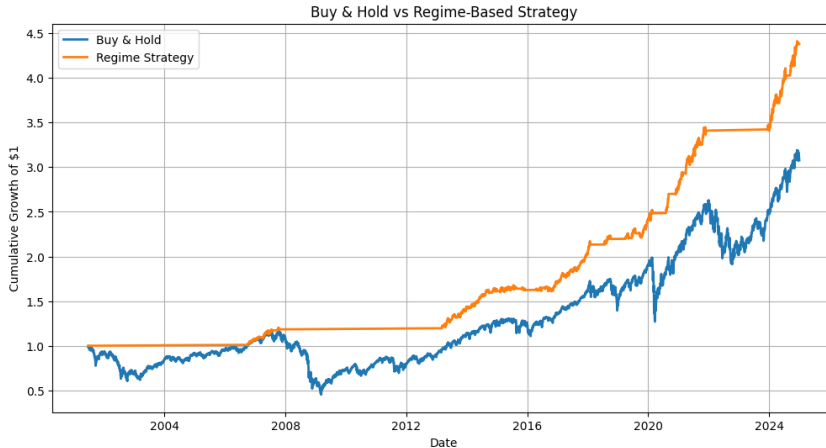
Metric	Buy & Hold	Regime Strategy
Total Return	207.8%	337.9%
Volatility	19.2%	4.9%
Sharpe Ratio	0.35	1.30
Max Drawdown	-61.0%	-4.5%

Key Result

Regime-based strategy outperforms in all aspects

Strategy Performance Comparison

Growth of \$1 Investment Over Time



Conclusion

Key Findings:

- Financial markets exhibit **distinct regimes**
- AI methods successfully detect hidden market states
- Market regimes show strong **persistence over time**

Main Outcome

Regime-based strategy improves returns while reducing risk

Final Insight: Adapting to market conditions is better than staying constant

References

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Thank You

Thank You

Questions & Discussion